

Two inputs.

① x_1 : hours studied.

② x_2 : hours slept.

$$Z = w_1 x_1 + w_2 x_2 + b$$

Activation function (take Sigmoid)

$$y = \frac{1}{1 + e^{-z}} \text{ (sigmoid)} \geq 0.5$$

$$< 0.5 \rightarrow 0$$

$$w_1 = 2, \quad w_2 = 1.5, \quad b = -15$$

$$x_1 = 6$$

$$x_2 = 5$$

$$Z = w_1(x_1) + w_2(x_2) + b$$

$$= 2(6) + (1.5)5 + 15 = 4.5$$

Sigmoid Probability.

$$y = \frac{1}{1 + e^{-z}} = \frac{1}{1 + e^{-4.5}} = 0.9$$

Pass ≥ 0.5 , fail < 0.5 = Pass.